Time—30 minutes

45 QUESTIONS

For each question in this section, choose the best answer and blacken the corresponding space on the answer sheet.

Each question below consists of a word in capital letters, followed by five lettered words or phrases. Choose the word or phrase that is most nearly opposite in meaning to the word in capital letters. Since some of the questions require you to distinguish fine shades of meaning, consider all the choices before deciding which is best.

Example:

GOOD: (A) sour (B) bad (C) red (D) hot (E) ugly

Ⓐ ● Ⓒ Ⓓ Ⓔ

1. CONTAMINATE: (A) activate (B) exhibit (C) diagnose (D) cleanse (E) generate

2. EXPLICIT: (A) illegal (B) fulfilled (C) flat (D) unclear (E) fortunate

3. INDUSTRIOUS: (A) sensitive (B) lazy (C) cruel (D) imitative (E) unpaid

4. DEPLETE: (A) disclose (B) surround (C) add to (D) vary from (E) respond to

5. MARRED: (A) overheated (B) enjoyable (C) perfect (D) comfortable (E) tart

6. CLAMOR: (A) temporary setback (B) orderly discussion (C) disguised sympathy (D) plain demeanor (E) fair judgment

7. CONCUR: (A) laugh (B) disagree (C) compensate (D) release (E) exceed

8. PLASTIC: (A) unavailable (B) jagged (C) receding (D) indivisible (E) rigid

9. FLEDGLING: (A) impostor (B) veteran (C) opponent (D) reluctant supporter (E) accurate reporter

10. AMBULATORY: (A) bedridden (B) hostile (C) contagious (D) frail (E) glum

11. BUTTRESS: (A) straighten (B) forward (C) undermine (D) show off (E) observe secretly

12. SACCHARINE: (A) caustic (B) allergic (C) magnetic (D) reticent (E) polluted

13. EXPEDITE: (A) absolve (B) encompass (C) alter radically (D) speak in anger (E) make more difficult

14. VENERATION: (A) intrusion (B) collaboration (C) irreverence (D) continence (E) inversion

15. DESICCATE: (A) darken (B) sedate (C) migrate (D) soak (E) erase

Each sentence below has one or two blanks, each blank indicating that something has been omitted. Beneath the sentence are five lettered words or sets of words. Choose the word or set of words that best fits the meaning of the sentence as a whole.

Example:

Although its publicity has been ---, the film itself is intelligent, well-acted, handsomely produced, and altogether ----.

(A) tasteless. .respectable (B) extensive. .moderate (C) sophisticated. .amateur (D) risqué. .crude (E) perfect. .spectacular

● Ⓑ Ⓒ Ⓓ Ⓔ

16. With all its ----, the journey had nonetheless been an extremely eventful and successful one.

(A) discomfort (B) adventure (C) glory (D) diligence (E) accomplishment

17. The reactor is a tool of enormous ----: its energy is turned to the service of many disciplines, from metallurgy to archaeology.

(A) versatility (B) disparity (C) vagueness (D) fragility (E) fluctuation

18. The ---- of music these days provides the exposure that enables any person to discover that he or she has an ---- for some sort of music.

(A) polyphony. .esteem
(B) enrichment. .articulation
(C) classicism. .aptitude
(D) vibrance. .antipathy
(E) accessibility. .appreciation

19. There is little chance of ---- in a static society in which all people think and live alike.

(A) consensus (B) boredom (C) regularity (D) tranquility (E) innovation

20. By comparing the actual architectural ---- of the Aztec pyramids with those reported in ancient Aztec documents, it is possible to ---- the general descriptive accuracy of these records.

(A) designs. .falsify (B) details. .assess (C) blueprints. .delay (D) devices. .initiate (E) dimensions. .ignore

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1

Each passage below is followed by questions based on its content. Answer all questions following a passage on the basis of what is stated or implied in that passage.

Usually the designation "father of modern biology" is given to William Harvey. In the early 1600's, Harvey observed that the beating heart expelled the blood within it. He then reasoned: if the heart contains two ounces of blood and beats sixty-five times each minute, then it must eject into the body over ten pounds of blood per minute. It had previously been thought that blood was derived from the food that is eaten. But one cannot imagine ten pounds of blood being formed anew each minute from the amount of food a person consumes. Reflection on this observation and simple deduction led Harvey to postulate that blood expelled by the heart must circulate through the body and return to the heart. He then performed experiments to investigate the hypothesis. He showed that obstruction of a vein causes pooling of blood on the side of the obstruction away from the heart. He showed that the bleeding arises from the nearest end to the heart of a severed artery and the farthest end of a severed vein. And he demonstrated with elegant simplicity the function of the venous valves, concerning which he wrote that "so provident a cause as nature had not so plac'd many valves without design."

Harvey's discovery of the circulation of blood was a monumental and far-reaching contribution to science. In the first place, it discredited the beliefs of fourteen centuries that the heart was not a muscular organ and that the blood passed through the septum between the right and left ventricles. In addition, Harvey knowingly or unknowingly used the scientific method in almost astoundingly modern fashion. And, finally, he bolstered his already incontrovertible claim to immortality by conceiving the heart as a pump, for in this notion Harvey heralded a new view concerning living organisms—though he did not appreciate the meaning of his own work on this score. But to his successors his work gave proof to the concept that a living organism could be viewed as a material machine. Descartes was quick to praise Harvey for "having broken the ice in this matter," and in his own discourses he relied heavily on the work of Harvey to illustrate the mechanical nature of living objects. Their only difference from man-made machines, he insisted, was in the degree of complexity. With these assertions, sudden new excitement entered the realm of biological thought. For it now seemed reasonable that if the living organism were a material mechanism, then it, too, could be investigated by the new method of science.

21. It can be inferred that Harvey is considered the "father of modern biology" because

(A) he was the first to document his findings
(B) his experiments could be easily duplicated
(C) everyone before him had thought of the heart as a muscular organ
(D) he was the first to describe with precision the position of the heart
(E) he introduced the concept of mechanism into biology

22. The author seems to regard Harvey's work as which of the following?

(A) Revolutionary (B) Hypothetical
(C) Controversial (D) Rudimentary
(E) Prosaic

23. With which of the following is the passage primarily concerned?

(A) Explaining the workings of the human heart and the blood system
(B) Tracing the development of biology from Harvey to Descartes
(C) Discussing early misconceptions about the heart and blood
(D) Describing the significance to biology of William Harvey's work
(E) Criticizing investigations made prior to the adoption of the new scientific method

24. It can be inferred that Harvey's observation of pooling of blood on the side of a vein obstruction away from the heart led him to the conclusion that

(A) the blood was returning to the heart
(B) blood was manufactured as it circulated
(C) blood was expelled from the heart with each beat
(D) the heart had to pump harder when the blood was far away from it
(E) previous calculations as to the amount of blood contained by the body were incorrect

25. It can be inferred from the passage that the concept of mechanism was important to the scientific method because it supported the idea that

(A) living organisms were part of a universal design
(B) scientists could work together, each one performing part of an experiment
(C) hypotheses must be based on experimental evidence
(D) the somewhat complicated method could lead to a simplistic theory of life
(E) organisms could be expected to function in understandable and regular ways

GO ON TO THE NEXT PAGE

The reading passages in this test are brief excerpts or adaptations of excerpts from published material. The ideas contained in them do not necessarily represent the opinions of the College Board or Educational Testing Service. To make the text suitable for testing purposes, we may in some cases have altered the style, contents, or point of view of the original.

Is it wicked to take pleasure in spring, and other seasonal changes? Is it politically reprehensible, while we are living under an imperfect social order, to point out that life is frequently more worth living because of a bluebird's song, a maple tree in October, or some other natural phenomenon that does not cost money and does not have what newspaper editors call a political angle? There is no doubt that many people think so. A favorable reference to "Nature" in one of my articles is likely to bring me abusive letters, and though the key word in these letters is usually "sentimental," two ideas seem to be mixed up in them. One is that any pleasure in the actual process of life encourages a sort of political quietism. People, so the thought runs, ought to be discontented; we ought to multiply our wants and not simply increase our enjoyment of the things we already have. The other idea is that this is the age of machines and that to dislike the machine, or even to want to limit its domination, is reactionary and slightly ridiculous. Love of nature, the argument goes, is a foible of urbanized people who have no notion what nature is really like; those who have to deal with the soil do not love it, and take at most a utilitarian interest in birds and flowers. To love the country one must live in the town, merely taking an occasional weekend ramble at the warmer times of the year.

History disproves this last notion. The other idea seems to me wrong in a subtler way. Certainly we ought to be discontented, we ought not simply to make the best of a bad job, and yet if we kill all pleasure in the actual process of life, what sort of future are we preparing for ourselves? If we cannot enjoy the return of spring, what will we do with the leisure that the machine will give us? If our economic and political problems are solved, life may become simpler instead of more complex, and the sort of pleasure one gets from finding the first violet will loom larger than the sort one gets from watching the Sunday football game on television. By retaining one's childhood love of trees, butterflies, and even toads, one makes a peaceful and decent future a little more probable. But by preaching the doctrine that only steel and concrete are to be admired, one makes it a little surer that human beings will have no outlet for their surplus energy except in hatred and leader-worship.

26. Which of the following statements best represents the author's response to the opening question?

(A) Yes, spring never gave a person freedom.
(B) Yes, the seasons are unimportant in a machine age.
(C) Yes, pleasure has no place in an unjust society.
(D) No, nature makes us forget our discontent.
(E) No, appreciation of nature makes life more bearable.

27. According to the passage, many readers find references to nature in the author's articles to be

(A) politically offensive
(B) delightfully sentimental
(C) morally admirable
(D) inadequately documented
(E) carelessly irrelevant

28. In the last sentence, the author states that exclusive admiration of "steel and concrete" may have which of the following effects?

(A) Alienation from the ideals of society
(B) Disappearance of parks and wooded areas
(C) Loss of sympathy for the perceptions of a child's mind
(D) Replacement of human affections with mechanical efficiency
(E) Diversion of natural impulses to harmful ends

29. The author answers his opponents by doing all of the following EXCEPT

(A) providing an explanation of his own views
(B) asking rhetorical questions
(C) refuting each argument exhaustively
(D) taking a longer view than they do
(E) claiming to understand their point of view

30. The author would most likely agree with which of the following statements?

(A) Financial price determines value.
(B) Life will become increasingly complex.
(C) Farmers have no affection for nature.
(D) Machines can serve a beneficial purpose.
(E) Childhood values do not suit adults.

GO ON TO THE NEXT PAGE

Select the word or set of words that best completes each of the following sentences.

31. All these plans and programs are going to produce no --- changes in our cities since they touch only the surface of the ugliness, the --- ways in which the system manifests its disorders.

(A) exterior. .bizarre
(B) minimal. .obnoxious
(C) appreciable. .vital
(D) visible. .petty
(E) significant. .superficial

32. His ---- disposition, by turns sunny and sullen, was a reflection not of inner turmoil but of external stimuli.

(A) somber (B) contentious (C) stoic
(D) volatile (E) callous

33. While regent of Japan, Hideyoshi desired approval of his poetry, and since there was no dearth of sycophants at the court, his work was almost unanimously ----.

(A) amended (B) distilled (C) interred
(D) collated (E) extolled

34. So great was her desire to explain every nuance of her metaphysical theory that even the variety and copiousness of her native language seemed ---- the ---- of that desire.

(A) inadequate to. .fervor
(B) untapped by. .intensity
(C) untouched by. .rigor
(D) sufficient for. .fanaticism
(E) unmatched by. .torpor

35. The present court is ----, for never before has partisan politics played such a demonstrable role in judicial decisions or so ---- the quality of jurisprudence.

(A) unique. .colored
(B) effective. .interrupted
(C) impartial. .reflected
(D) perfect. .investigated
(E) didactic. .complicated

Each question below consists of a related pair of words or phrases, followed by five lettered pairs of words or phrases. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair.

Example:

YAWN : BOREDOM :: (A) dream : sleep
(B) anger : madness (C) smile : amusement
(D) face : expression (E) impatience : rebellion

Ⓐ Ⓑ ● Ⓓ Ⓔ

36. FISH : OCEAN :: (A) snake : path (B) bird : cage
(C) bear : zoo (D) camel : desert
(E) elephant : river

37. PAINTER : SMOCK :: (A) mason : trowel
(B) lumberjack : ax (C) baby : bib
(D) umpire : baseball (E) prophet : beard

38. SOB : GRIEF :: (A) apology : punishment
(B) letter : acceptance (C) whisper : silence
(D) cheer : approval (E) voice : music

39. LOTTERY : DRAWING :: (A) gambling : dice
(B) election : tally (C) education : grades
(D) football : practice (E) hockey : penalties

40. AVIARY : FLY :: (A) aquarium : swim
(B) kennel : climb (C) solarium : observe
(D) satellite : launch (E) terrarium : glide

41. CRESCENDO : INTENSITY ::
(A) innocence : youth (B) acceleration : speed
(C) uproar : attention (D) adaptation : power
(E) abbreviation : length

42. CONFLUENCE : STREAMS ::
(A) greenhouse : plants (B) foundation : buildings
(C) resemblance : pictures (D) junction : roads
(E) ebb : tides

43. FLIPPANT : DEFERENTIAL ::
(A) vain : handsome (B) frivolous : serious
(C) devout : sincere (D) studious : intelligent
(E) indifferent : apathetic

44. CONJUGAL : MARRIAGE ::
(A) spiritual : sin (B) corrupt : politics
(C) fiscal : money (D) deteriorating : alliance
(E) blissful : retirement

45. PERFIDY : TRAITOR ::
(A) execution : criminal (B) loyalty : contributor
(C) veracity : patriot (D) generosity : benefactor
(E) pacifism : belligerent

S T O P

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY.
DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

SECTION 2

Time—30 minutes

25 QUESTIONS

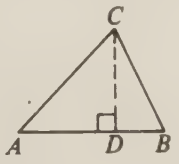
In this section solve each problem, using any available space on the page for scratchwork. Then decide which is the best of the choices given and blacken the corresponding space on the answer sheet.

The following information is for your reference in solving some of the problems.

Circle of radius r: Area $= \pi r^2$; Circumference $= 2\pi r$
The number of degrees of arc in a circle is 360.
The measure in degrees of a straight angle is 180.

Definitions of symbols:

$=$	is equal to	$\leq$	is less than or equal to
$\neq$	is unequal to	$\geq$	is greater than or equal to
$<$	is less than	$\parallel$	is parallel to
$>$	is greater than	$\perp$	is perpendicular to



Triangle: The sum of the measures in degrees of the angles of a triangle is 180.
If $\angle CDA$ is a right angle, then

(1) area of $\triangle ABC = \dfrac{AB \times CD}{2}$

(2) $AC^2 = AD^2 + DC^2$

Note: Figures which accompany problems in this test are intended to provide information useful in solving the problems. They are drawn as accurately as possible EXCEPT when it is stated in a specific problem that its figure is not drawn to scale. All figures lie in a plane unless otherwise indicated. All numbers used are real numbers.

---

1. Jane had 4.25 meters of rope and used 3.5 meters. How many meters of rope does she have left?

   (A) 0.25 (B) 0.75 (C) 1.25
   (D) 1.75 (E) 7.75

2. Which of the following is NOT equal to a whole number?

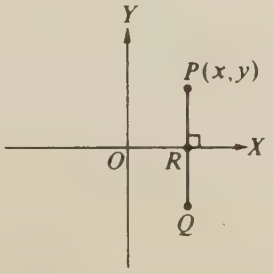
   (A) $\dfrac{54+27}{3}$ (B) $\dfrac{54+27}{9}$ (C) $\dfrac{54+27}{18}$
   (D) $\dfrac{54+27}{27}$ (E) $\dfrac{54+27}{81}$

3. If the price of nails has been increased from 5 pounds for \$1 to 3 pounds for \$1, how many fewer pounds can now be bought for \$6 than could be bought before?

   (A) 2
   (B) 3
   (C) 8
   (D) 12
   (E) 15

4. The expression "2 less than the product 3 times x" can be written as

   (A) $\dfrac{3x}{2}$
   (B) $2 - 3x$
   (C) $3x - 2$
   (D) $(3 - 2)x$
   (E) $2 + 3x$



5. If $PR = RQ$ in the figure above, what are the coordinates of point Q?

   (A) $(x, -y)$ (B) (y, x) (C) $(y, -x)$
   (D) $(-x, y)$ (E) $(-x, -y)$

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6. If $2x = -1$, then $(2x - 3)^2 =$

(A) 0 (B) 1 (C) 4 (D) 16 (E) 25

7. If $9 \times 9 \times 9 = \frac{27 \times 27}{r}$, then $r =$

(A) $\frac{1}{3}$

(B) 1

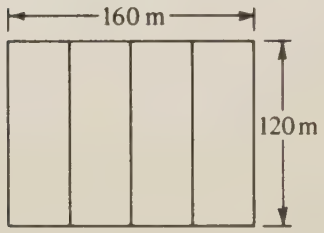
(C) 3

(D) 6

(E) 9

8. $0.84^2 - 0.83^2 =$

(A) 0.00167
(B) 0.0167
(C) 0.167
(D) 16.70
(E) 167.0



9. A gardener wants to enclose a rectangular lot and divide it into four equal parts, as shown above. If every line segment represents one section of fence and no sections of fence overlap, how many meters of fencing are needed?

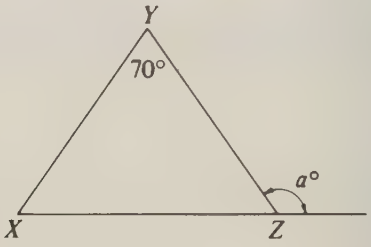
(A) 560 (B) 760 (C) 800
(D) 920 (E) 980

10. The average (arithmetic mean) of a student's scores on four tests was 78. If she received a score of 70 on each of the first two tests and 84 on the third, what was her score on the fourth test?

(A) 58
(B) 69
(C) 77
(D) 80
(E) 88

11. If $x - y = 8$, what is the value of $x^2 - y^2$?

(A) −64 (B) −16 (C) 16 (D) 64
(E) It cannot be determined from the information given.



12. In $\triangle XYZ$ above, if $XY = YZ$, then $a =$

(A) 125 (B) 115 (C) 110 (D) 70 (E) 55

13. The fraction $\frac{45}{60}$ equals all of the following EXCEPT

(A) $\frac{3}{4}$ (B) $\frac{6}{8}$ (C) $\frac{9}{12}$ (D) $\frac{12}{15}$ (E) $\frac{15}{20}$

14. If $ax + y = x + 2y = 7$ and $x = 1$, then $a =$

(A) 2
(B) 3
(C) 4
(D) 5
(E) 6

GO ON TO THE NEXT PAGE

15. If the distances between points A, B, and C are equal, which of the following could be true?

I. A, B, and C are vertices of an equilateral triangle.
II. A, B, and C are vertices of a right triangle.
III. B and C lie on an arc of a circle with center A.

(A) I only (B) II only (C) III only
(D) I and III (E) II and III

16. Let k be any integer. If $\vec{k} = 2k$ and $\overleftarrow{k} = \frac{1}{2}k$, what does the product $\vec{k}$ times $\overleftarrow{k}$ equal?

(A) 1 (B) 4 (C) $\frac{k^2}{2}$ (D) k (E) k^2

17. What fraction of 15 hours is 15 seconds?

(A) $\frac{1}{60}$

(B) $\frac{1}{225}$

(C) $\frac{1}{360}$

(D) $\frac{1}{3{,}600}$

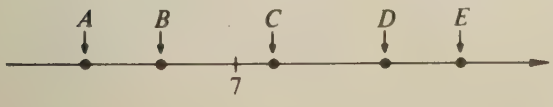
(E) $\frac{1}{54{,}000}$

Questions 18-19 refer to the following definition.

The d-distance of an integer M from an integer N is $M - N$.

18. What is the d-distance of 5 from -5 ?

(A) -10 (B) -5 (C) 0 (D) 5 (E) 10



19. Of the integers indicated on the number line above, which has the least d-distance from 7 ?

(A) A (B) B (C) C (D) D (E) E

20. At a certain college, x liters of milk are needed per month for each student. At this rate, y liters of milk will supply z students for how many months?

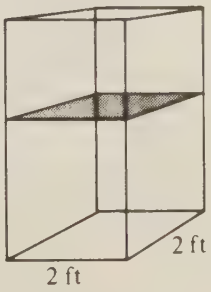
(A) $\frac{y}{xz}$

(B) $\frac{z}{xy}$

(C) $\frac{xy}{z}$

(D) $\frac{yz}{x}$

(E) xyz



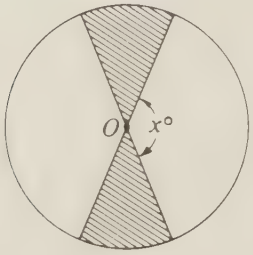
21. The figure above shows water in a tank whose base is 2 feet by 2 feet. When a solid cube 1 foot on an edge is completely immersed in the water, how many inches will the water rise in the tank?

(A) $\frac{1}{4}$ (B) 3 (C) 4 (D) 12 (E) 24

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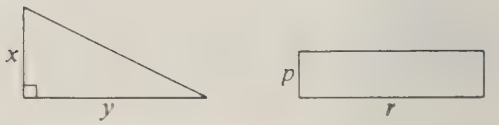
22. N years ago Sue was twice as old as Joe was. If Joe is now 25 years old, how old is Sue now in terms of N?

(A) $50 - 2N$
(B) $50 - N$
(C) $25 - N$
(D) $25 + N$
(E) $50 + N$



23. The circle above has center O and radius 2. If the total area of the shaded regions formed by the two diameters is π, then $x =$

(A) 45 (B) 90 (C) 120 (D) 135 (E) 270



24. If the triangle and rectangle above have equal areas and if $\frac{pr}{2} = 60$, then $xy =$

(A) 240 (B) 120 (C) 60 (D) 30 (E) 15

---

25. If the sum of the first n positive integers is x, then, in terms of n and x, which of the following equals the sum of the next n integers?

(A) nx
(B) $n + x$
(C) $n^2 + x$
(D) $2n + x$
(E) $n + 2x$

S T O P

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY.
DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

SECTION 3

Time—30 minutes

40 QUESTIONS

For each question in this section, choose the best answer and blacken the corresponding space on the answer sheet.

Each question below consists of a word in capital letters, followed by five lettered words or phrases. Choose the word or phrase that is most nearly opposite in meaning to the word in capital letters. Since some of the questions require you to distinguish fine shades of meaning, consider all the choices before deciding which is best.

Example:

GOOD: (A) sour (B) bad (C) red (D) hot (E) ugly

Ⓐ ● Ⓒ Ⓓ Ⓔ

1. MINGLE: (A) revolve (B) despise (C) flow smoothly (D) remain apart (E) regulate well

2. VALID: (A) unjustified (B) impervious (C) unrepentant (D) inevitable (E) unclaimed

3. EGG ON: (A) remove (B) restrain (C) respect (D) conclude (E) immerse

4. CONVENTIONAL: (A) ornamental (B) unorthodox (C) misunderstood (D) widely dispersed (E) rapidly constructed

5. OPTIMUM: (A) most expensive (B) most diverse (C) least precise (D) least familiar (E) least favorable

6. SAGE: (A) bystander (B) procrastinator (C) fanatic (D) eccentric (E) ignoramus

7. VOUCHSAFE: (A) isolate (B) weaken (C) refuse to grant (D) talk crudely (E) abstain from pleasure

8. CLEMENCY: (A) remorse (B) severity (C) incompetence (D) private opinion (E) careful consideration

9. INTRACTABLE: (A) obvious (B) talented (C) bountiful (D) obedient (E) joyous

10. SUFFRAGE: (A) lack of eloquence (B) lack of franchise (C) lack of pain (D) desire for power (E) desire for retribution

Each sentence below has one or two blanks, each blank indicating that something has been omitted. Beneath the sentence are five lettered words or sets of words. Choose the word or set of words that best fits the meaning of the sentence as a whole.

Example:

Although its publicity has been ---, the film itself is intelligent, well-acted, handsomely produced, and altogether ----.

(A) tasteless. .respectable (B) extensive. .moderate (C) sophisticated. .amateur (D) risqué. .crude (E) perfect. .spectacular

● Ⓑ Ⓒ Ⓓ Ⓔ

11. Only since the 1850's have large numbers of women been able to ---- the scientific community in the United States, and even today they have to ---- lingering male bias.

(A) rescind. .submit to
(B) penetrate. .protect
(C) escape. .combat
(D) enter. .contend with
(E) evaluate. .seek out

12. The attempt to demonstrate that matter can exist in a form too small to be seen by the eye revealed that it is possible to discover truth ----, that is, by a process of inference based on observation.

(A) immediately (B) indirectly (C) momentarily (D) infallibly (E) definitively

13. We cannot endorse developments that make our surroundings more hideous, our culture more ----, or our lives less complete.

(A) viable (B) effectual (C) tolerant (D) provocative (E) tawdry

GO ON TO THE NEXT PAGE

14. Her political thought was developed in a series of essays characterized by a reliance on ----; generalizations and imprecise grand statements --- her.

(A) universalities. .evaded
(B) minutiae. .disturbed
(C) details. .preoccupied
(D) abstractions. .disconcerted
(E) ambiguities. .infuriated

15. Unfortunately, routine military operations are described ---- in this account of the conflict, whereas analyses of the more significant political considerations are relegated to ---- paragraphs.

(A) concisely. .lavish
(B) summarily. .interminable
(C) exhaustively. .occasional
(D) superficially. .brief
(E) adequately. .ample

Each question below consists of a related pair of words or phrases, followed by five lettered pairs of words or phrases. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair.

Example:

YAWN : BOREDOM :: (A) dream : sleep (B) anger : madness (C) smile : amusement (D) face : expression (E) impatience : rebellion

Ⓐ Ⓑ ● Ⓓ Ⓔ

16. GLAZE : CLAY :: (A) easel : paint (B) color : chalk (C) varnish : wood (D) title : book (E) tea : cup

17. FLEET : SHIPS :: (A) dictionary : syllables (B) committee : people (C) clock : time (D) photograph : subjects (E) hospital : nurses

18. THREATEN : ATTACK :: (A) cancel : revise (B) annihilate : defile (C) rehabilitate : discard (D) promise : deliver (E) assume : doubt

19. SILKWORM : SILK :: (A) oyster : pearl (B) branch : leaf (C) ant : anthill (D) moth : wool (E) mosquito : disease

20. SONATA : MUSICAL :: (A) epic : whimsical (B) novel : literary (C) song : humorous (D) testimony : rhymed (E) pantomime : vocal

21. CENTRIFUGE : SEPARATION ::
(A) dial : instrumentation
(B) degree : measurement
(C) thermometer : refrigeration
(D) apparatus : filtration
(E) microscope : magnification

22. CATACOMB : CEMETERY :: (A) subway : railroad (B) dirigible : airplane (C) elevator : shaft (D) bridge : highway (E) grotto : cave

23. RED HERRING : MISLEAD ::
(A) smoke screen : obscure
(B) blank check : bewilder
(C) clean slate : condemn
(D) white lie : damage
(E) loan shark : haggle

24. APERTURE : CAMERA :: (A) rope : pulley (B) window : room (C) piston : engine (D) eyeglasses : vision (E) wrench : pipe

25. PERCIPIENT : DISCERNMENT ::
(A) ravenous : hunger (B) punctual : tact (C) heroic : conflict (D) artificial : reality (E) concerned : indifference

GO ON TO THE NEXT PAGE

Each passage below is followed by questions based on its content. Answer all questions following a passage on the basis of what is stated or implied in that passage.

Despite their seemingly infinite formulations, invisible inks generally fall into one of three categories, depending on the way they are made visible.

Line
(5) First there are organic liquids such as milk, lemon juice, vinegar, and some body fluids. These inks are crude and are detected by heating the paper to which they are applied. Gentle warming actually chars the inks into visibility. Although they can be detected easily, organic inks have the advantage of
(10) being always available.

The second category of inks comprises chemical solutions. Colorless when dry, they become visible when treated with an appropriate reagent. Such inks are called "sympathetic inks" because each writing
(15) solution requires a specific developing reagent to make the message appear. Of course, the trick with sympathetic inks is to find compounds that are extremely reagent-specific or, better yet, that must be developed in several intermediate steps before
(20) color appears.

A third type of ink—radioactive ink—is developed by photographic processes. Solutions of radium D in acetone, for example, have been used to write messages on clothing without raising the suspicion of
(25) counterespionage agents who were looking for secret ink on paper. Messages written in this way are developed simply by exposing the clothes to an X-ray film plate.

It is also possible to write messages with plain
(30) water and avoid the use of chemicals altogether. Water disturbs paper fibers; the message can be made visible by exposure to iodine vapor. The vapor settles on the paper's surface in the written lines where the fibers have been disturbed.

26. According to the passage, "sympathetic inks" (line 14) are so named because they

(A) have an affinity for various types of writing surfaces
(B) resemble each other in that they leave no color trace when dry
(C) are developed through interaction with a particular substance
(D) can be made visible only for a limited length of time
(E) possess a great degree of internal molecular cohesion

27. It can be inferred from the passage that compounds "that must be developed in several intermediate steps" (lines 18-19) are desirable particularly because

(A) their color definition is improved proportionately
(B) interception of messages is more difficult
(C) complex procedures themselves encourage further technical innovation
(D) legibility increases as an ink requires more developing reagents
(E) such processes facilitate the timely detection of any error

28. It is most likely that the author does not list water as a fourth category of invisible writing fluids because it

(A) does not need a distinct developing process
(B) is so easily made visible as to be of little use
(C) has not yet gained acceptability as an espionage tool
(D) is quickly noticeable even to ordinary citizens
(E) is too elementary a substance to be considered an ink

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3

My first glimpse of the flat black stretches of Chicago depressed and dismayed me, mocked all my fantasies. Chicago seemed an unreal city whose mythical houses were built of slabs of black coal wreathed in palls of gray smoke, houses whose foundations were sinking slowly into the dank prairie. Flashes of steam showed intermittently on the wide horizon, gleaming translucently in the winter sun. The din of the city entered my consciousness, entered to remain for years to come. The year was 1927.

What would happen to me here? Would I survive? My expectations were modest. I wanted only a job. Hunger had long been my daily companion. Diversion and recreation, with the exception of reading, were unknown. In all my life—though surrounded by many people—I had not had a single satisfying, sustained relationship with another human being and, not having had any, I did not miss it. I made no demands whatever upon others.

The train rolled into the depot. Aunt Maggie and I got off and walked slowly through the crowds into the station. I looked about to see if there were signs saying: FOR WHITE—FOR COLORED. I saw none. Black people and White people moved about, each seemingly intent upon a private mission. There was no racial fear. And yet, because everything was so new, I began to grow tense again, although it was a different sort of tension than I had known before. I knew that this machine-city was governed by strange laws and I wondered if I would ever learn them.

As I stood in the icy wind waiting for the streetcar, I wanted to talk to Aunt Maggie, to ask her questions, but her tight face made me hold my tongue. I was learning already from the frantic light in her eyes the strain that the city imposed upon its people. I was seized by doubt. Should I have come here? But going back was impossible. I had fled a known terror, and perhaps I could cope with this unknown terror that lay ahead.

The streetcar came. Aunt Maggie motioned for me to get on and pushed me toward a seat in which a White man sat looking blankly out the window. I sat down beside the man and looked straight ahead of me. After a moment I stole a glance at the White man; he was still staring out the window, his mind fastened upon some inward thought. I did not exist for him; I was as far from his mind as the stone buildings that swept past in the street. It would have been illegal for me to sit beside him in the part of the South that I had come from.

29. The passage as a whole is best characterized as
    (A) a description of an individual's reactions to a new environment
    (B) a portrait of the residents of a city
    (C) an illustration of the energy and optimism of youth
    (D) an analysis of racial discrimination in an urban setting
    (E) a comparison between romantic illusions and mundane reality

30. In the context of the passage, the author's "fantasies" (line 3) most probably have to do with
    (A) dreams of great personal wealth and power
    (B) memories of scenes and characters from favorite books
    (C) theories about the nature of personal relationships
    (D) the desire for physical isolation and intellectual independence
    (E) notions of the magnificence of the city

31. To the author, Chicago seemed to be all of the following EXCEPT
    (A) boring
    (B) gloomy
    (C) crowded
    (D) noisy
    (E) unpredictable

32. The author's attitude in lines 38-41 is best described as
    (A) deep despair
    (B) pretended indifference
    (C) calm assurance
    (D) apprehensive resolve
    (E) excited expectation

33. On the basis of the passage, which of the following statements about Aunt Maggie can most logically be made?
    (A) She has just had an argument with the author.
    (B) She will find the author a job.
    (C) She has become indifferent to the welfare of the author.
    (D) She wishes she were anywhere but Chicago.
    (E) She feels the pressure of life in Chicago.

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The chief reason for the development of the medieval Italian into the modern Italian lies in the character of the states of Renaissance Italy including, ironically, even the despotisms. In the Middle Ages both sides of human consciousness—that which was turned within and that which was turned without—lay as though dreaming or half awake beneath a common veil. The veil was woven of faith, illusion, and childlike prepossession, through which the world and history were seen clad in strange hues. Medieval people were conscious of themselves only as members of a race, people, party, family, or corporation—only through some general category. It was in Italy that the veil was first lifted. In Italy there arose an objective treatment and consideration of the state and of all things of this world, and at the same time a subjective side asserted itself with corresponding emphasis. Italians began to recognize themselves as individuals, and they began to behave in accordance with this recognition.

Despotism fostered the utmost individuality not only in the tyrant but also in the persons who were protected or used by the tyrant—the secretaries, ministers, poets, and companions. These people were forced to know all the inward resources of their own nature, the momentary as well as the permanent. Their enjoyment was enhanced and concentrated by the desire to obtain the greatest satisfaction from a period of relative power and influence that held no guarantee of long duration.

34. According to the author, which of the following was a paradox of Renaissance Italy?

(A) Italy was the birthplace of individualism.
(B) Renaissance people perceived the state objectively.
(C) Renaissance people recognized themselves as spiritual individuals.
(D) Renaissance people gained more power as they became more independent.
(E) Despotism helped lead to a greater sense of individuality.

35. Which of the following best describes the style of lines 4-11 ?

(A) Rambling (B) Satirical (C) Ironic (D) Metaphoric (E) Paradoxical

36. It can be inferred from the passage that those people "who were protected or used by the tyrant" (lines 23-24) were NOT likely to believe that they

(A) should be able to depend on themselves when the need arose
(B) were always in danger of being deprived of their power or influence
(C) ought to make the most of any good fortune that befell them
(D) had identities independent of any group allegiance
(E) could disregard the demands and authority of the despot

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Confronted by an enigmatic spectacle the writer descends within the self, and in that lonely inward region of stress and strife the writer, if deserving and fortunate, finds the terms of his or her appeal. The appeal is made to our less obvious capacities: to that part of our nature which, because of the warlike conditions of existence, is necessarily kept out of sight within the more resisting and hard qualities—like the vulnerable body within a steel armor. The appeal is less loud, more profound, less distinct, more stirring—and sooner forgotten. Yet its effect endures forever. The changing wisdom of successive generations discards ideas, questions facts, demolishes theories. But the writer appeals to that part of our being which is not dependent on wisdom; to that in us which is a gift and not an acquisition—and, therefore, more permanently enduring. The writer speaks to our capacity for delight and wonder, to the sense of mystery surrounding our lives; to our sense of pity, and beauty, and pain; to the latent feeling of fellowship with all creation—and to the subtle but invincible conviction of solidarity that knits together the loneliness of innumerable hearts, to the solidarity in dreams, in joy, in sorrow, in aspirations, in illusions, in hope, in fear, which binds people to each other, which binds together all humanity—the dead to the living and the living to the yet unborn.

Line
(5)
(10)
(15)
(20)
(25)

37. According to the passage, if writers are to find the terms of their "appeal" (line 4), they must be which of the following?

I. Clever
II. Worthy
III. Lucky

(A) I only (B) III only (C) I and II only
(D) II and III only (E) I, II, and III

38. The author suggests that a writer appeals to a "part of our nature" (line 6) that is

(A) unsympathetic (B) analytical
(C) innate (D) readily apparent
(E) hard and resisting

39. It can be inferred that the author would most likely describe "wisdom" (lines 12 and 15) as

(A) preeminent (B) mutable (C) timeless
(D) liberating (E) accumulative

40. The author's primary point about writers is that they

(A) descend within themselves to avoid confrontation with the mass of humanity
(B) have an effect that is permanently enduring because they appeal to the wisdom of humanity
(C) address the underlying sense of affinity with others that exists within all human beings
(D) choose to ignore the joys, dreams, sorrows, and fears of other human beings
(E) can foretell what existence will be like for future generations of humans

S T O P

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY.
DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

SECTION 5

Time—30 minutes

35 QUESTIONS

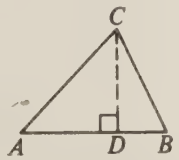
In this section solve each problem, using any available space on the page for scratchwork. Then decide which is the best of the choices given and blacken the corresponding space on the answer sheet.

The following information is for your reference in solving some of the problems.

Circle of radius r: Area $= \pi r^2$; Circumference $= 2\pi r$
The number of degrees of arc in a circle is 360.
The measure in degrees of a straight angle is 180.

Definitions of symbols:

$=$	is equal to	$\leqq$	is less than or equal to
$\neq$	is unequal to	$\geqq$	is greater than or equal to
$<$	is less than	$\parallel$	is parallel to
$>$	is greater than	$\perp$	is perpendicular to



Triangle: The sum of the measures in degrees of the angles of a triangle is 180.
If $\angle CDA$ is a right angle, then

(1) area of $\triangle ABC = \frac{AB \times CD}{2}$

(2) $AC^2 = AD^2 + DC^2$

Note: Figures which accompany problems in this test are intended to provide information useful in solving the problems. They are drawn as accurately as possible EXCEPT when it is stated in a specific problem that its figure is not drawn to scale. All figures lie in a plane unless otherwise indicated. All numbers used are real numbers.

---

1. If $a + 2 = 4$, then $2a + 2 =$

   (A) 4 (B) 6 (C) 8 (D) 10 (E) 12

Tree	Height (in centimeters)
Cherry	170
Peach	190
Apple	205
Plum	185
Pear	210

2. Exactly how many of the trees listed in the table above are more than 2 meters high? (1 meter = 100 centimeters)

   (A) One (B) Two (C) Three
   (D) Four (E) Five

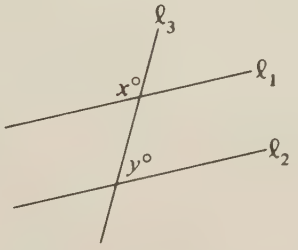
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3. $2x(3x + 5y) =$

   (A) $5x + 7xy$ (B) $6x + 10xy$ (C) $6x^2 + 5y$
   (D) $6x^2 + 10y$ (E) $6x^2 + 10xy$

4. Kate and Beth are both members of team X, which played n games in a certain season. Kate played in $\frac{1}{2}$ of the games and Beth played in 12 of the games. If Beth played in fewer games than Kate, which of the following is a possible value for n?

   (A) 10 (B) 14 (C) 18 (D) 24 (E) 26



5. If parallel lines ℓ_1 and ℓ_2 are intersected by line ℓ_3, what is the value of $x + y$?

   (A) 180 (B) 150 (C) 120 (D) 90
   (E) It cannot be determined from the information given.

---

6. $\frac{27}{1000} + \frac{5}{100} + \frac{3}{10} =$

   (A) 0.0035 (B) 0.2753 (C) 0.3527
   (D) 0.377 (E) 0.72

7. A traffic flow of 1,440 cars during a 24-hour period is equivalent to a rate of how many cars per minute?

   (A) 240 (B) 120 (C) 60 (D) 24 (E) 1

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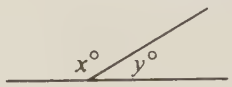
5

Questions 8-27 each consist of two quantities, one in Column A and one in Column B. You are to compare the two quantities and on the answer sheet blacken space

A if the quantity in Column A is greater;
B if the quantity in Column B is greater;
C if the two quantities are equal;
D if the relationship cannot be determined from the information given.

Notes: 1. In certain questions, information concerning one or both of the quantities to be compared is centered above the two columns.
2. In a given question, a symbol that appears in both columns represents the same thing in Column A as it does in Column B.
3. Letters such as x, n, and k stand for real numbers.

EXAMPLES

	Column A	Column B	Answers
E1.	2×6	$2 + 6$	● Ⓑ Ⓒ Ⓓ
E2.	 $180 - x$	y	Ⓐ Ⓑ ● Ⓓ
E3.	$p - q$	$q - p$	Ⓐ Ⓑ Ⓒ ●

	Column A	Column B
8.	$2^2 + 2$	2^3

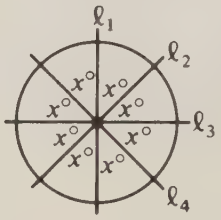
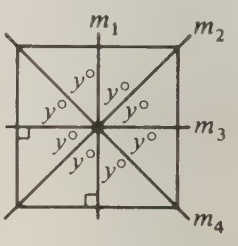
The total price of 1 apple and 1 peach is 75 cents.

9.	The price of 1 apple	The price of 1 peach
10.	$\frac{1}{\frac{5}{6}}$	1.6

x and y are both positive integers.

$xy = 30$

11.	x	y

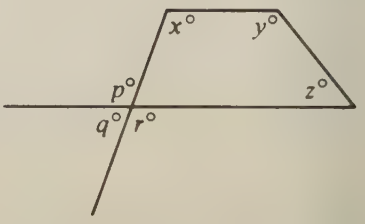
	Column A	Column B
12.	 The value of x	 The value of y

Car X travels at an average speed of 60 kilometers per hour.

13.	The distance traveled by car X in $\frac{3}{4}$ hour at its average speed	40 kilometers

$x - y > 0$

14.	x	y

15.	 $p + q + r$	$x + y + z$

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SUMMARY DIRECTIONS FOR COMPARISON QUESTIONS

Answer: A if the quantity in Column A is greater;
B if the quantity in Column B is greater;
C if the two quantities are equal;
D if the relationship cannot be determined from the information given.

	Column A	Column B
	Last year the gross income of Corporation C was \$5,200,000. This year its gross income was 10 percent higher than last year.	
16.	The gross income of Corporation C this year	\$5,700,000

	$t < -1$	
17.	$(t-1)^2(t+1)^2$	0

	$\triangle ABC$ is equilateral.	
18.	Length of the altitude to side AB	Length of side AB



	The sum of the numbers in the row is equal to the sum of the numbers in the column.	
19.	x	y

	$5x + 5y = 30$	
20.	The average (arithmetic mean) of x and y	3

	A wheel with a radius of 50 centimeters rolls without slipping along a straight line and completes one revolution each second.	
21.	Distance along the line that the wheel travels in 2 seconds	400 centimeters

	Column A	Column B
	x is a prime number and $21 < x < 30$.	
22.	x	24
23.	The sum of two different odd positive integers each less than 10	The sum of three different even positive integers each less than 10
	$2x + y = 7$ $x + 2y = 2$	
24.	$x + y$	3
25.	Area of the triangular region bounded by the X-axis, Y-axis, and the graph of the line $x + y = 6$	18
	The operation $\triangledown$ is defined for all nonzero numbers x and y by the equation $x \triangledown y = \left(\frac{x}{y}\right)^2$.	
	$wz \neq 0$	
26.	$w \triangledown z$	$w \triangledown (-z)$
	a and b are positive integers.	
27.	$\frac{a}{b}$	$\frac{a+1}{b+1}$

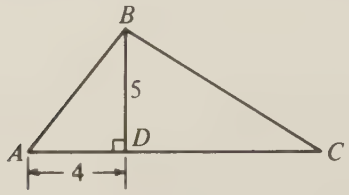
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Solve each of the remaining problems in this section using any available space for scratchwork. Then decide which is the best of the choices given and blacken the corresponding space on the answer sheet.

28. If each time the symbol $\sim$ is used it can represent either of the arithmetic operations $+$ or $\times$, which of the following could be the result of $1 \sim (2 \sim 4)$? (In this expression, the second $\sim$ need not represent the same operation as does the first $\sim$).

I. 7
II. 8
III. 9

(A) I only
(B) II only
(C) I and III only
(D) II and III only
(E) I, II, and III

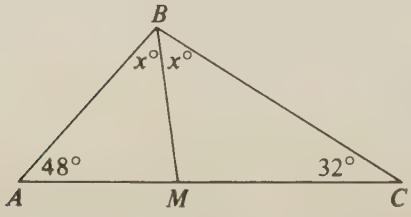


29. In the figure above, if the area of $\triangle ABC$ is 30, then the area of $\triangle BDC$ is

(A) 5 (B) 10 (C) 15 (D) 20 (E) 40

30. What is the thickness, in centimeters, of 1 sheet of paper if the thickness of a uniform pack of 500 such sheets is 2.5 centimeters?

(A) 0.005 (B) 0.02 (C) 0.05
(D) 0.125 (E) 0.2



31. In $\triangle ABC$ above, what is the value of x?

(A) 40 (B) 42 (C) 50 (D) 58
(E) It cannot be determined from the information given.

32. If 12 and 16 each divide N without remainder, what must be the value of N?

(A) 24 (B) 48 (C) 120 (D) 192
(E) It cannot be determined from the information given.

33. In the Lost Dog Kennel there is a total of 25 dogs. Of these, exactly 10 dogs have black spots, exactly 12 dogs have brown spots, and exactly 5 dogs have both brown and black spots. What is the total number of dogs in the kennel that have neither brown spots nor black spots?

(A) 2
(B) 3
(C) 8
(D) 13
(E) 15

34. The number of boys attending Fairfield High School is twice the number of girls. If $\frac{1}{6}$ of the boys and $\frac{1}{4}$ of the girls are in the school band, what fraction of the students at Fairfield are in the school band?

(A) $\frac{5}{36}$
(B) $\frac{7}{36}$
(C) $\frac{2}{9}$
(D) $\frac{7}{24}$
(E) $\frac{5}{12}$

35. A square floor with sides of length 6 meters is to be tiled with square tiles. If each tile has a perimeter of 1 meter, what is the minimum number of such tiles needed to cover the floor?

(A) 12 (B) 24 (C) 36 (D) 144 (E) 576

S T O P

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY.
DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

Correct Answers for Scholastic Aptitude Test
Form Code 3F

VERBAL		MATHEMATICAL	
Section 1	Section 3	Section 2	Section 5
1. D	1. D	1. B	1. B
2. D	2. A	2. C	2. B
3. B	3. B	3. D	3. E
4. C	4. B	4. C	4. E
5. C	5. E	5. A	5. A
6. B	6. E	6. D	6. D
7. B	7. C	7. B	7. E
8. E	8. B	8. B	*8. B
9. B	9. D	9. D	*9. D
10. A	10. B	10. E	*10. B
11. C	11. D	11. E	*11. D
12. A	12. B	12. A	*12. C
13. E	13. E	13. D	*13. A
14. C	14. B	14. C	*14. A
15. D	15. C	15. D	*15. C
16. A	16. C	16. E	*16. A
17. A	17. B	17. D	*17. A
18. E	18. D	18. E	*18. B
19. E	19. A	19. A	*19. B
20. B	20. B	20. A	*20. C
21. E	21. E	21. B	*21. A
22. A	22. A	22. B	*22. D
23. D	23. A	23. D	*23. D
24. A	24. B	24. A	*24. C
25. E	25. A	25. C	*25. C
26. E	26. C		*26. C
27. A	27. B		*27. D
28. E	28. E		28. E
29. C	29. A		29. D
30. D	30. E		30. A
31. E	31. A		31. C
32. D	32. D		32. E
33. E	33. E		33. C
34. A	34. E		34. B
35. A	35. D		35. E
36. D	36. E		
37. C	37. D		
38. D	38. C		
39. B	39. B		
40. A	40. C		
41. B			
42. D			
43. B			
44. C			
45. D			

*Indicates four-choice questions. (All of the other questions are five-choice.)

The Scoring Process

Machine-scoring is done in three steps:

- *Scanning.* Your answer sheet is "read" by a scanning machine and the oval you filled in for each question is recorded on a computer tape.
- *Scoring.* The computer compares the oval filled in for each question with the correct response. Each correct answer receives one point; omitted questions do not count toward your score. For each wrong answer, a fraction of a point is subtracted to correct for random guessing. For questions with five answer choices, one-fourth of a point is subtracted for each wrong response; for questions with four answer choices, one-third of a point is subtracted for each wrong response. The SAT-verbal test has 85 questions with five answer choices each. If, for example, a student has 44 right, 32 wrong, and 9 omitted, the resulting raw score is determined as follows:

$$44 \text{ right} - \frac{32 \text{ wrong}}{4} = 44 - 8 = 36 \text{ raw score points}$$

  Obtaining raw scores frequently involves the rounding of fractional numbers to the nearest whole number. For example, a raw score of 36.25 is rounded to 36, the nearest whole number. A raw score of 36.50 is rounded upward to 37.

- *Converting to reported scaled score.* Raw test scores are then placed on the College Board scale of 200 to 800 through a process that adjusts scores to account for minor differences in difficulty among different editions of the test. This process, known as equating, is performed so that a student's reported score is not affected by the edition of the test taken nor by the abilities of the group with whom the student takes the test. As a result of placing SAT scores on the College Board scale, scores earned by students at different times can be compared. For example, an SAT-verbal score of 400 on a test taken at one administration indicates the same level of developed verbal ability as a 400 score obtained on a different edition of the test taken at another time.

How to Score the Test

You can verify the College Board SAT scores reported to you recently by using the information in this booklet along with the copy of your answer sheet. *Before you begin, check that the first two characters (number and letter) of the form code you marked in item 3 on your answer sheet are the same as the form code printed on the front of this booklet.* Compare the responses shown on the copy of your answer sheet with the list of correct answers.

SAT-Verbal Sections 1 and 3

Step A: Count the number of correct answers for *section 1* and record the number in the space provided on the worksheet on the next page. Then do the same for the incorrect answers. (Do not count omitted answers.) To determine subtotal A, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal A}$$

Step B: Count the number of correct answers and the number of incorrect answers for *section 3* and record the numbers in the spaces provided on the worksheet. To determine subtotal B, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal B}$$

Step C: To obtain C, add subtotal A to subtotal B, keeping any decimals. Enter the resulting figure on the worksheet.

Step D: To obtain D, your raw verbal score, round C to the nearest whole number. (For example, any number from 44.50 to 45.49 rounds to 45.) Enter the resulting figure on the worksheet.

Step E: To find your reported SAT-verbal score, look up the total raw verbal score you obtained in step D in the conversion table on the back cover. Enter this figure on the worksheet. (The SAT-verbal score you just recorded and your reported SAT-verbal score should be identical. If not, see the paragraph at the bottom of the next page.)

SAT-Mathematical Sections 2 and 5

Step A: Count the number of correct answers and the number of incorrect answers for *section 2* and record the numbers in the spaces provided on the worksheet. To determine the subtotal A, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal A}$$

Step B: Count the number of correct answers and the number of incorrect answers for the *five-choice questions (questions 1 through 7 and 28 through 35) in section 5* and record the numbers in the spaces provided on the worksheet. To determine the subtotal B, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal B}$$

Step C: Count the number of correct answers and the number of incorrect answers for the *four-choice questions (questions 8 through 27) in section 5* and record the numbers in the spaces provided on the worksheet. To determine the subtotal C, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{3} = \text{subtotal C}$$

Step D: To obtain D, add subtotal A, subtotal B, and subtotal C, keeping any decimals. Enter the resulting figure on the worksheet.

Step E: To obtain E, your raw mathematical score, round D to the nearest whole number. (For example, any number from 44.50 to 45.49 rounds to 45.) Enter the resulting figure on the worksheet.

Step F: To find your reported SAT-mathematical score, look up the total raw mathematical score you obtained in E in the conversion table on the back cover. Enter this figure on the worksheet. (The SAT-mathematical score you just recorded and your reported SAT-mathematical score should be identical. If not, see the paragraph at the bottom of the next page.)

SAT-SCORING WORKSHEET

FORM CODE 3F

SAT-Verbal Sections

A. Section 1: _____ (no. correct) − ¼ (_____) (no. incorrect) = _____ subtotal A

B. Section 3: _____ (no. correct) − ¼ (_____) (no. incorrect) = _____ subtotal B

C. Total unrounded raw score (Total A + B) _____ C

D. Total rounded raw score (Rounded to nearest whole number) _____ D

E. SAT-verbal reported scaled score (See the conversion table on the back cover.) [ ] SAT-verbal score

SAT-Mathematical Sections

A. Section 2: _____ (no. correct) − ¼ (_____) (no. incorrect) = _____ subtotal A

B. Section 5: Questions 1 through 7 and 28 through 35 (5-choice) _____ (no. correct) − ¼ (_____) (no. incorrect) = _____ subtotal B

C. Section 5: Questions 8 through 27 (4-choice) _____ (no. correct) − ⅓ (_____) (no. incorrect) = _____ subtotal C

D. Total unrounded raw score (Total A + B + C) _____ D

E. Total rounded raw score (Rounded to nearest whole number) _____ E

F. SAT-mathematical reported scaled score (See the conversion table on the back cover.) [ ] SAT-math score

Should you have any questions on these scoring instructions, you may call the phone number below. If, after following the above scoring directions and checking your work at least twice, your results disagree with the SAT-verbal or SAT-mathematical score reported on your ATP score report, you may request rescoring of your answer sheet. If rescoring confirms that an error had been made (resulting in either higher or lower scores than those originally reported), corrected reports will be sent to all recipients of your original scores. Please send your request to:

College Board ATP
Box 592
Princeton, NJ 08541
Attention: Rescore Request
Telephone: (609) 771-7600

Please indicate whether it is your SAT-verbal or SAT-mathematical score, or both, that you want to be rescored. When you write, please include a copy of this scoring worksheet on which you did your calculations.

Score Conversion Table
Scholastic Aptitude Test
Form Code 3F

Raw Score	College Board Reported Score		Raw Score	College Board Reported Score	
	SAT-Verbal	SAT-Math		SAT-Verbal	SAT-Math
85	800		40	450	610
84	780		39	440	600
83	770		38	440	590
82	760		37	430	580
81	750		36	420	570
80	740		35	420	560
79	730		34	410	550
78	720		33	410	540
77	710		32	400	530
76	700		31	390	520
75	700		30	390	510
74	690		29	380	500
73	680		28	370	490
72	670		27	370	480
71	670		26	360	470
70	660		25	350	460
69	650		24	350	450
68	640		23	340	440
67	640		22	330	430
66	630		21	330	430
65	620		20	320	420
64	610		19	310	410
63	600		18	310	400
62	600		17	300	390
61	590		16	290	380
60	580	800	15	290	370
59	580	780	14	280	360
58	570	770	13	270	360
57	560	760	12	270	350
56	550	750	11	260	340
55	550	740	10	250	330
54	540	730	9	250	320
53	530	720	8	240	320
52	530	710	7	230	310
51	520	710	6	230	300
50	520	700	5	220	290
49	510	690	4	210	290
48	500	680	3	210	280
47	500	670	2	200	270
46	490	660	1	200	260
45	480	650	0	200	250
44	480	640	– 1	200	240
43	470	630	– 2	200	240
42	460	620	– 3	200	230
41	460	610	– 4	200	220
			– 5	200	210
			– 6 or below	200	200